

Turn this assignment in on gradescope. All work must be neat and legible. Turn in a “final draft”. There should be nothing scratched out and your work should flow generally from left to right and top to bottom.

(Yes - there are only two problems!)

1. Find two non-trivial solutions to $x^2 - 60y^2 = 1$. (Use the Chakravala algorithm; you don't need to show the arithmetic, but you do need to show the steps in the algorithm. That is, the (x, y, k) in each step).

Since at each step of the algorithm, we want to minimize $|m^2 - N|$, we start with $m = 8$ since $8^2 = 64$ is closest to 60, and get the first triple $(8, 1, 4)$. For our first improvement, we need to find the next m such that $\frac{x+ym}{k}$ is an integer, while still minimizing $|m^2 - 60|$. We see that $m \equiv 0 \pmod{4}$, and again, we have $m = 8$ is the closest to $\sqrt{60}$. Our next x is

$$x_{\text{new}} = \frac{xm + 60y}{|k|} = \frac{8 * 8 + 60 * 1}{4} = 31.$$

Our next y is

$$y_{\text{new}} = \frac{x + ym}{|k|} = \frac{8 + 1 * 8}{4} = 4.$$

Our next k is

$$k_{\text{new}} = \frac{m^2 - 60}{k} = \frac{64 - 60}{4} = 1.$$

Thus, we have found one solution, namely $(31, 4, 1)$. Now, we can take advantage of Brahmagupta's composition operator to find another solution.

$$(31, 4, 1) \oplus (31, 4, 1) = (31 * 31 + 60 * 4 * 4, 31 * 4 + 4 * 31, 1 * 1) = (1921, 248, 1).$$

2. Using the algorithm with starting values $(10, 1, 8)$ find a solution to $x^2 - 92y^2 = 1$. Once you get to the point where $k = \pm 4$, use some reduction to get to a solution (note: I showed how to do this when $k = \pm 2$; do something similar here). Show your work as I mentioned above.

Instead of writing out all the arithmetic, I will list out each step in the following table.

x	y	k	m
10	1	8	6
19	2	-7	8
48	5	4	-

We will now try something similar to method used for $k = \pm 2$. First we use Brahmagupta's composition operator on the triple with itself:

$$(48, 5, 4) \oplus (48, 5, 4) = (48^2 + 92 \cdot 5^2, 2 \cdot 48 \cdot 5, 16) = (4604, 480, 16).$$

This gives us the equation

$$4604^2 - 92 \cdot 480^2 = 16.$$

Notice that both 4604 and 480 are divisible by 4, so we can divide the entire equation by 16 to get

$$1151^2 - 92 \cdot 120^2 = 1.$$

Thus, we have found a solution, namely $(1151, 120, 1)$.